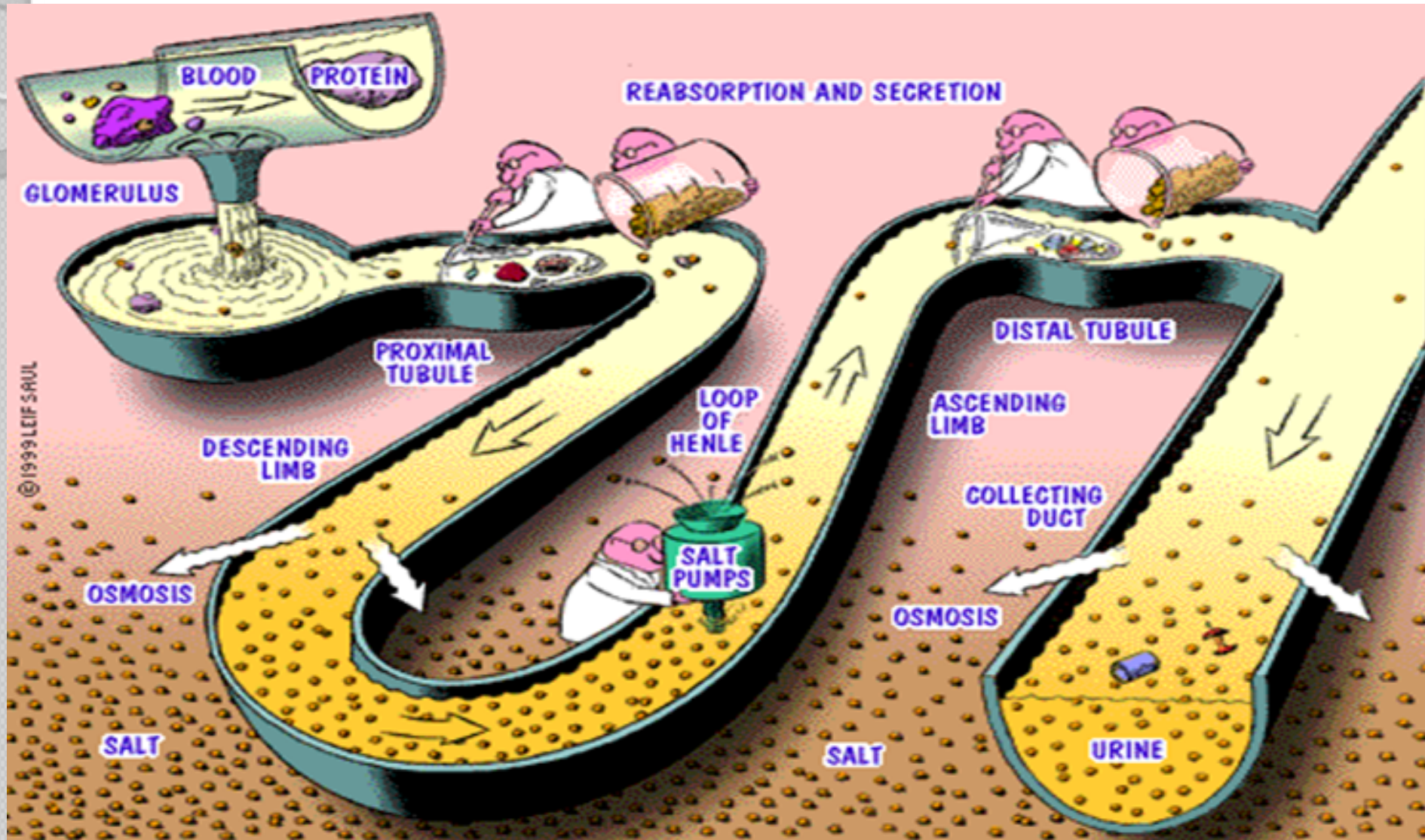


Tubular function



Objectives :

- 1) General principles of renal tubular transport
- 2) sodium handling .
- 3) potassium handling.
- 4) water handling.
- 5) glucose handling.
- 6) diuretics.



Main tubular function

Reabsorption (R)

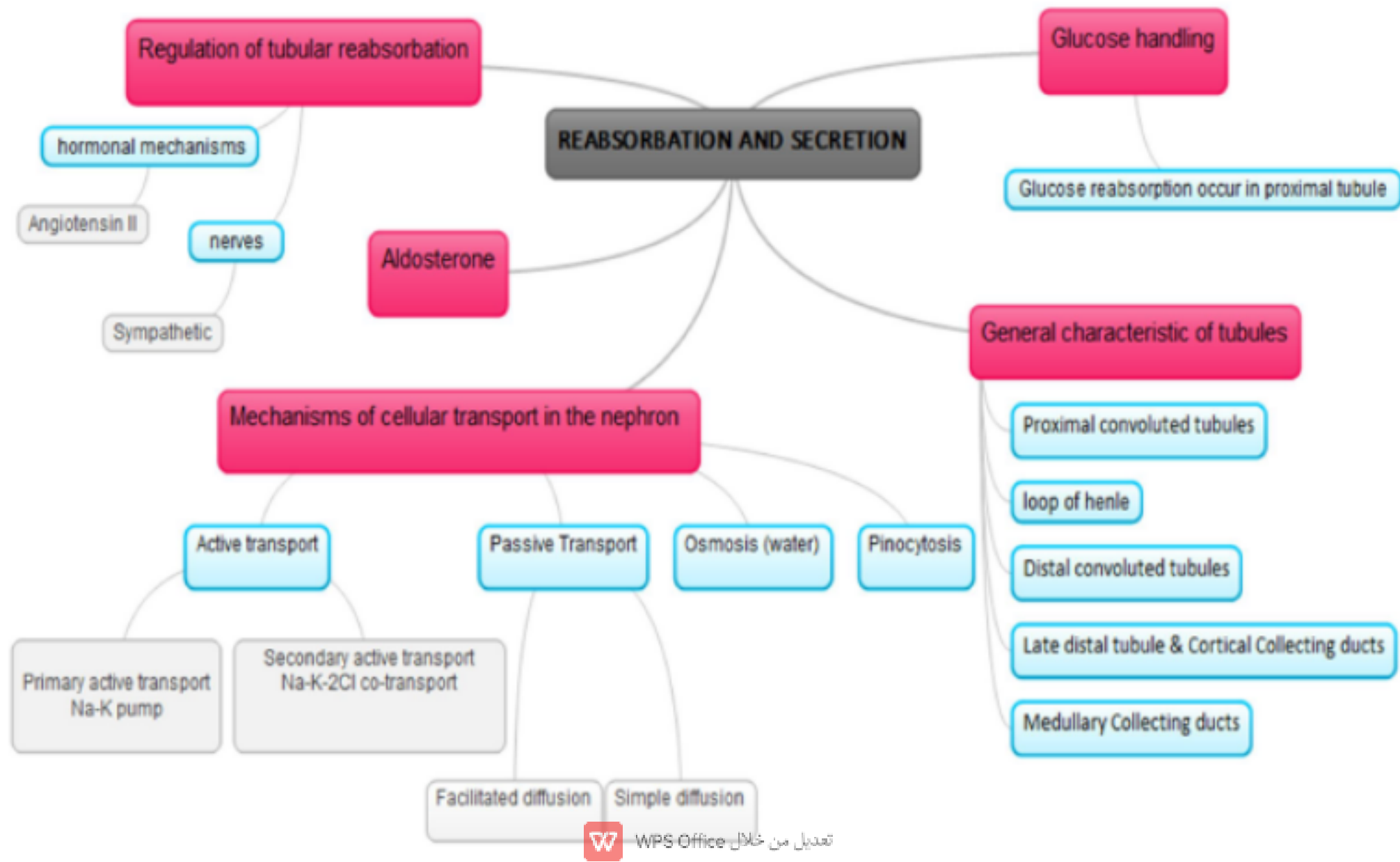
- is the transport of substance from the tubular lumen (filtrate) to the peritubular capillary (plasma).

Secretion (S)

- is the transport of substance from the peritubular capillary (plasma) to the tubular lumen (filtrate)

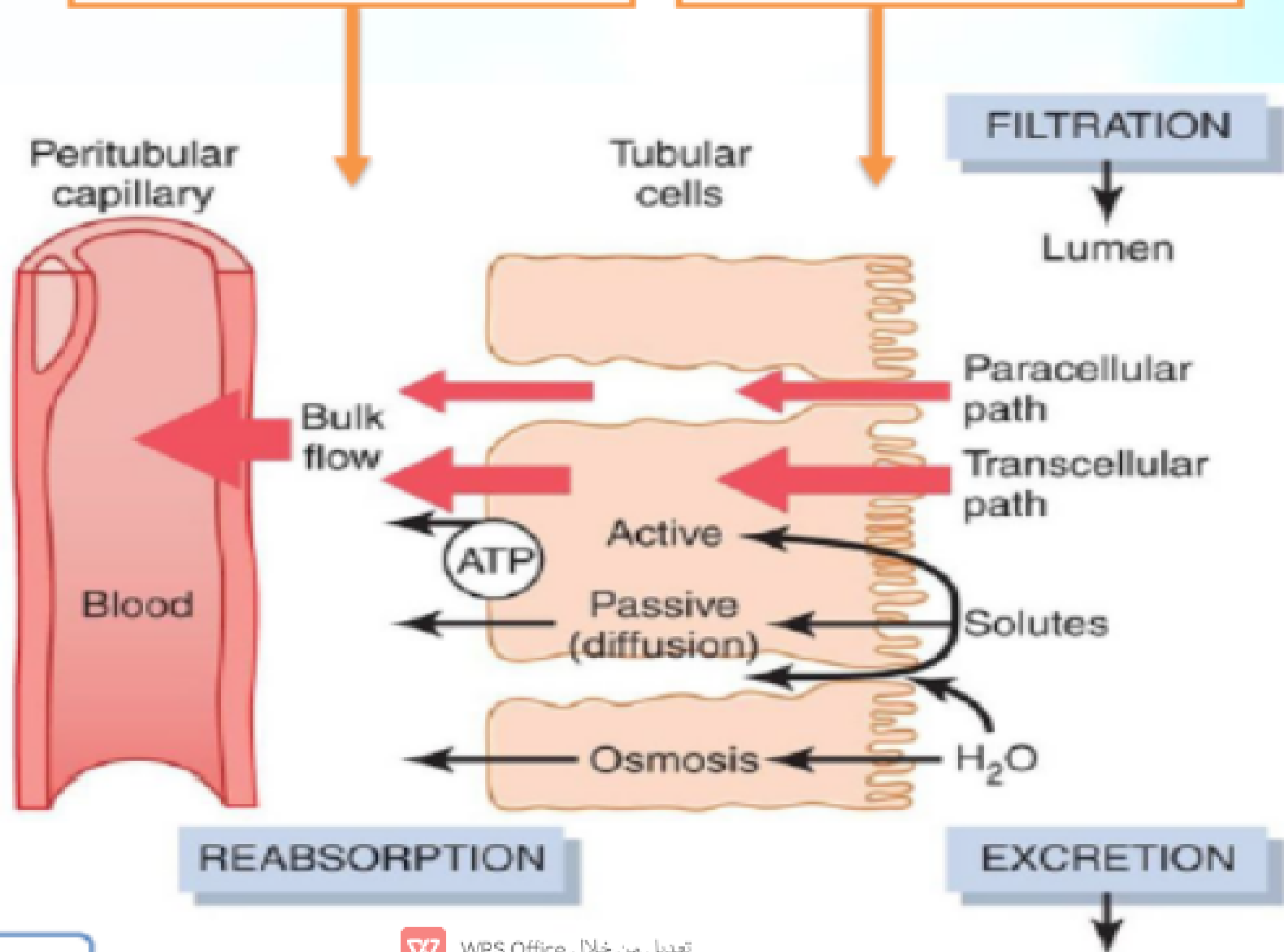


Mind Map



Second, through the renal interstitium, and back into the blood through the Peritubular capillary membrane

First, Reabsorption of filtered water and solutes from the tubular lumen across the tubular epithelial cells



GENERAL PRINCIPLES OF RENAL TUBULAR TRANSPORT

Transport mechanisms across cell membrane

1) Passive transport

- i. Diffusion
- ii. Facilitated diffusion (carrier, coupled transport , uniport or symport)
- iii. Solvent drag.

2) Active transport

3) Trans-epithelial transport

4) Para-cellular transport



Mechanisms of cellular transport in the nephron are:

Active transport

"Active transport can move a solute against an electrochemical gradient and requires energy derived from metabolism"

Primary active transport

Transport that is coupled directly to an energy source such as ATP

Sodium-potassium pump

(found in basolateral membrane along renal tubules)

H⁺-pump

Secondary active transport

Transport that is coupled indirectly to an energy source due to concentration gradient of ion

Na-K-2Cl co-transport

glucose-sodium co-transport (SGLT)

amino acid-sodium co-transport

H⁺/Na counter-transport

Passive Transport

Simple diffusion

(without carrier protein)

Cl, HCO₃⁻, urea, creatinine

Facilitated diffusion

(require carrier protein)

Glucose and amino acids at the basolateral border (GLUT)

Osmosis

Water is always reabsorbed by a passive (nonactive) physical mechanism called osmosis, which means water diffusion from a region of low solute concentration (high water concentration) to one of high solute concentration (low water concentration).

General characteristic of tubules

Proximal convoluted tubule

1- Proximal tubules is **coarse adjustment** (reabsorption a most of of water and solutes)

3-100% of glucose and amino acids reabsorbed

2- Solute reabsorption in the proximal tubule is isosmotic (equal amount of solute and water are reabsorbed)



Loop of henle

1- Responsible for producing a concentrated urine in the medulla.

2- When ADH (antidiuretic hormone) is present, water is reabsorbed and urine is concentrated.

Descending limb: (concentrate urine "1")

- 1- water permeable and allow absorption of 25% of filtered H₂O.
- 2- It is impermeable to Na-CL.
- 3- fluid become hyper-osmolar

Thin ascending limb: (dilute urine "2")

- 1- impermeable to H₂O
- 2- permeable to NaCl (passive)
- 3- By the end of ascending limb of loop, the tubular fluid becomes hypo-osmolar

Thick ascending limb: (dilute urine)

- 1- impermeable to H₂O
- 2- Na-K-2Cl co-transport occur in this part. (active)
- 3- the tubular fluid becomes hypo-osmolar to plasma in this part

(1) Concentrate urine = remove water from fluid

(2) Dilute urine = remove solutes from urine

* The main function of tubules is **concentrate urine** and that has done in **loop of henle of Juxtamedullary nephron**

* **Sodium-potassium pump** that found in **distal convoluted tubules** is under control of **aldosterone**

Distal convoluted tubules

1- Distal tubule is **fine adjustment**
(reabsorption a fine amount of water and solutes by hormonal control based on body needs)

2- The first portion of DCT forms part of Juxtaglomerular Apparatus, that provides feedback control of GFR and RBF of the same nephron.

3- The next early portion has the same characteristics as ascending limb of Henle that is

- 1- impermeable to water
- 2- absorbs solutes.

So it is called the **diluting segment** & the osmotic pressure of the fluid ~ 100 mOsm/L.



Late distal convoluted tubules & cortical collecting ducts

1- Composed of two types of cells:

- a. **Principal cells**: absorb Na^+ & H_2O and secrete K^+
- b. **Intercalated cells**: absorb K^+ & secrete H^+

• Secretion of K^+ and reabsorption of Na^+ controlled by aldosterone.

2- water permeability under ADH control. (works under body needs)

3- Impermeable to Urea.



Medullary collecting ducts

1- Under ADH control. (works under body needs)

2- Highly permeable to urea.

3- Final site for processing urine.

4- Secretes H^+ helps to:

a- maintain blood pH

b- reabsorb HCO_3^- and generate new HCO_3^-



Continues in renal pelvis,
ureters, urinary bladder
and urethra



Aldosterone

Function

- 1-increases Sodium reabsorption
- 2-stimulates Potassium secretion

When does it secreted?

- (1) Increased extracellular potassium concentration.
- (2) Increased angiotensin II levels, which typically occur in conditions associated with **sodium and volume depletion or low blood pressure (so it will increase blood pressure)**

Site of secretion

- Aldosterone, secreted by the **zona glomerulosa cells** of the **adrenal cortex**.

Mechanism of action

- by stimulating the sodium-potassium ATPase pump on the **basolateral side of the cortical collecting tubule membrane**.
- Aldosterone also increases the sodium permeability of the luminal side of the membrane .

Diseases associated with aldosterone

- Absence of aldosterone, as occurs with adrenal destruction or malfunction (**Addison's disease**)
- Excess aldosterone, as occurs in patients with adrenal tumors (**Conn's syndrome**) is associated with:
 - 1- sodium retention
 - 2- decreased plasma potassium concentration



	Reabsorption	Secretion
Proximal convoluted tubules	- Poorly reabsorbed of creatinine and urea - 60% - 75% of sodium and water - 90% of HCO_3, K^+, Ca and Cl^- - 100% Glucose and amino acids	- H^+ - Urea - ammonia
Descending loop of henle	-25% of water (H_2O)	_____
Thin ascending loop of henle	- Sodium chloride (NaCl)	_____
Thick ascending loop of henle	- Sodium - Potassium - Chloride	_____
Distal convoluted tubules	- Sodium in response to aldosterone - Water in response of ADH - Calcium in response of parathyroid hormone	- Potassium in response of aldosterone
Late distal tubule & Cortical Collecting ducts	a. Principal cells: absorb Sodium Na^+ & H_2O b. Intercalated cells: absorb Potassium K^+ & HCO_3 -Water in response of ADH - Sodium in response to aldosterone - Calcium in response of parathyroid hormone	a. Principal cells: secrete K^+ b. Intercalated cells: secrete H^+ Potassium in response of aldosterone
Medullary Collecting ducts	-Water in response of ADH - Highly permeable to urea (to maintain osmolarity of medulla)	- H^+



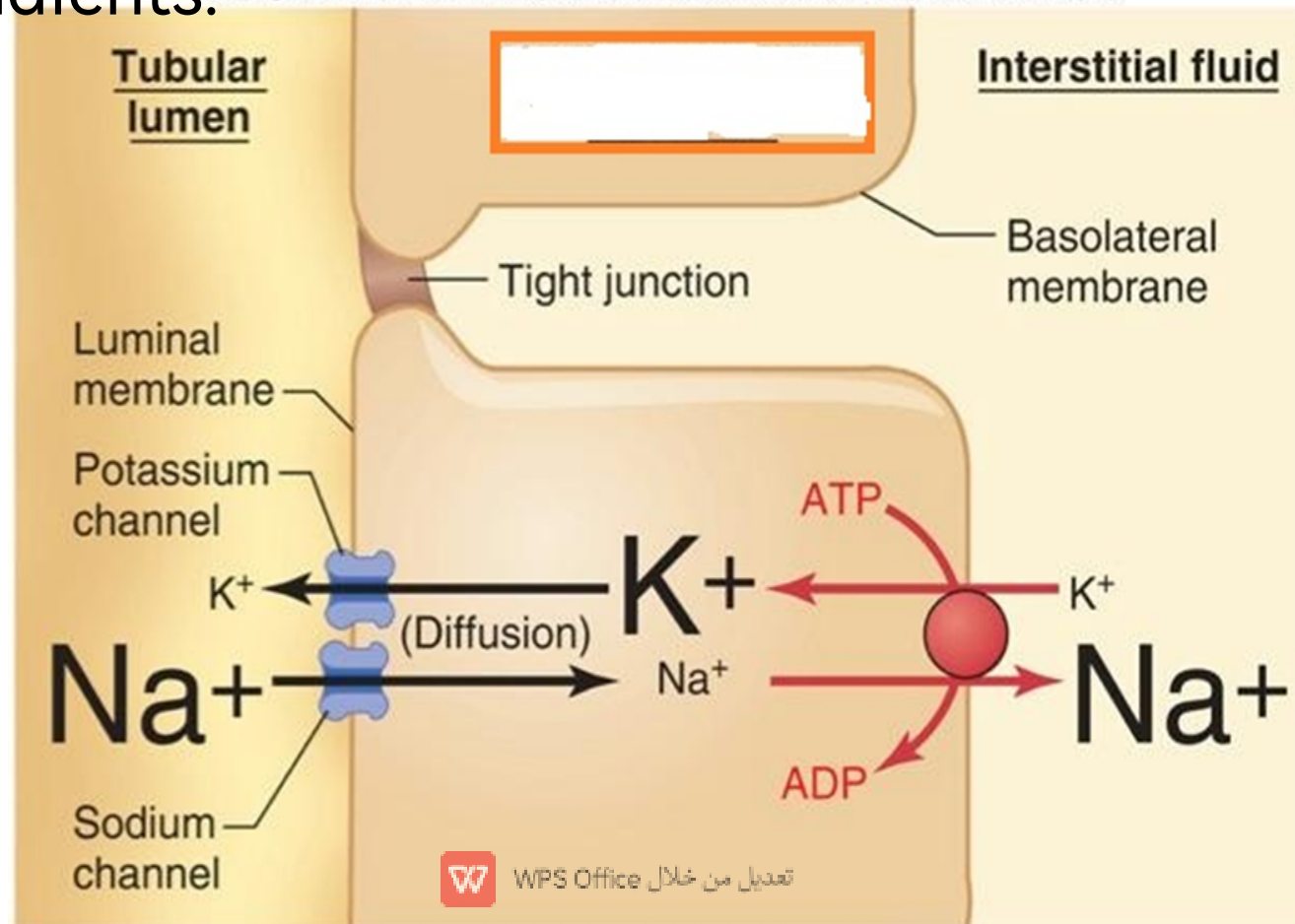
Sodium handling

- Na Reabsorption:
 - Na reabsorption plays a major role in body electrolyte and water metabolism.
- Some substances are co-transported with Na:
- Glucose
- Amino acids
- Organic acids
- Phosphate
- H⁺ (antiport)



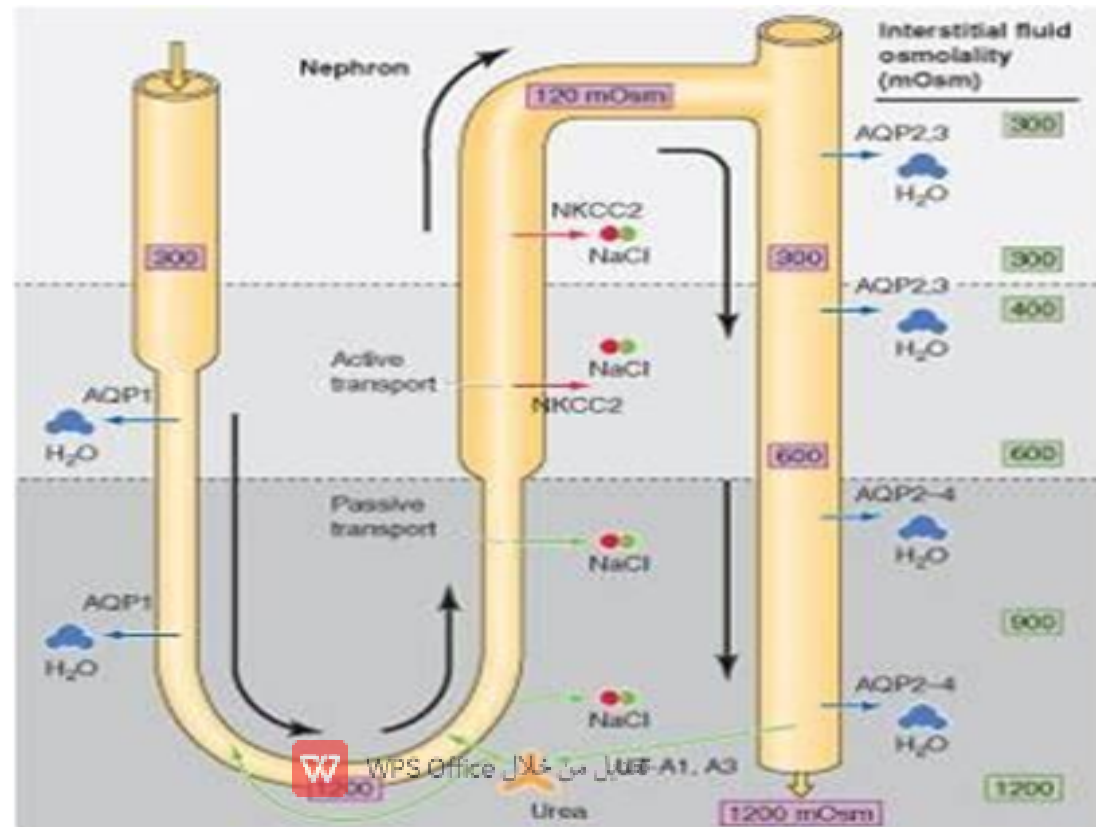
P.C.T:

- Reabsorbs 60–70% of the filtered Na.
- Reabsorption occurs actively by Na-K pump on the basolateral membrane.
- This allows passive Na diffusion through the luminal membrane down its chemical and electrical gradients.



LOOP OF HENLE:

- Reabsorbs 30% of the filtered Na .
- The thin descending limb does not reabsorb Na.
- The thin ascending limb reabsorbs Na passively.
- The thick ascending limb reabsorbs Na actively by Na-K-2CL co-transporter.



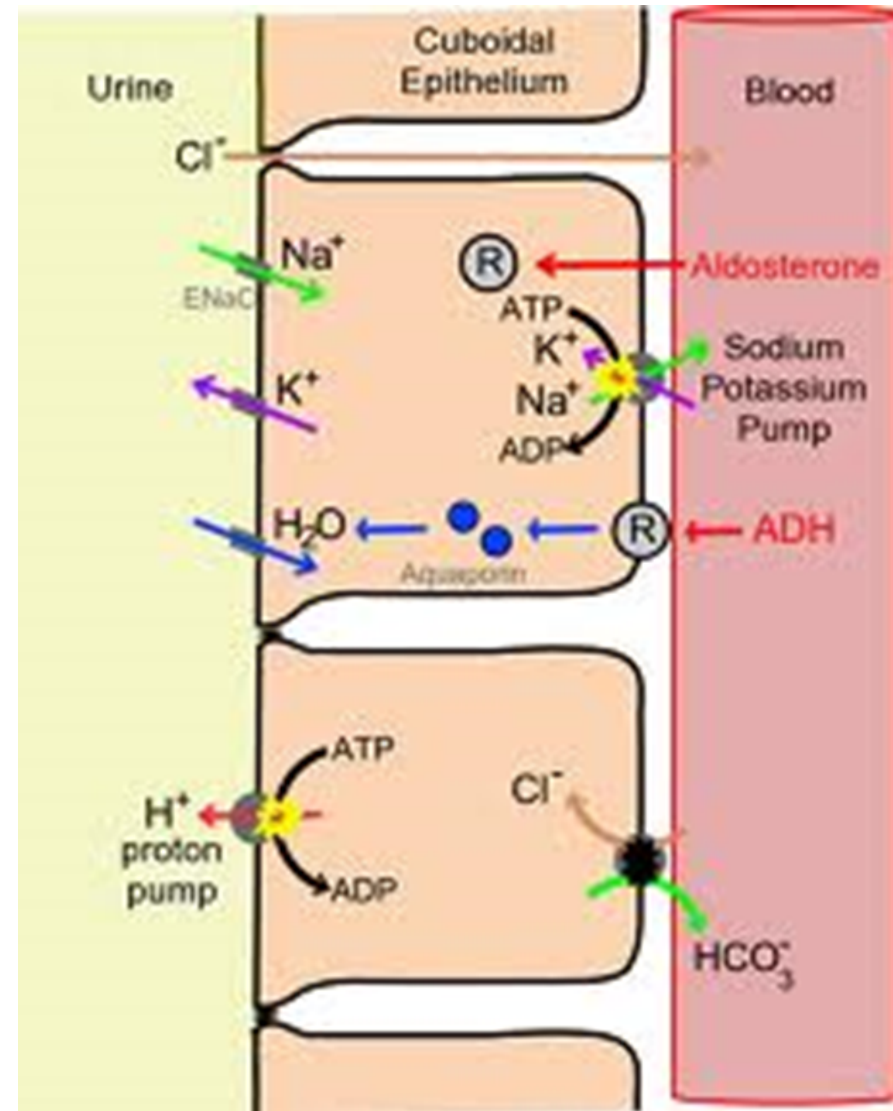
DCT & CD:

DCT

- About 7% of the filtered Na is reabsorbed by Na-Cl co transporter.

CD

- About 3% is absorbed via the ENaC channels, and this is the portion that is regulated by aldosterone.



Na excretion

- Normally, 96–99% of the filtered Na^+ is reabsorbed.
- The amount of Na^+ excreted is adjusted to equal the amount ingested (Na^+ balance). Thus urinary Na^+ output ranges from less than 1meq/d on low salt diet to 400meq/d when dietary Na^+ intake is high.
- Variation in Na^+ excretion are brought about by changes in GFR and Tubular reabsorption, primarily filtered Na^+ that reaches the collecting duct.



Potassium handling

PCT:

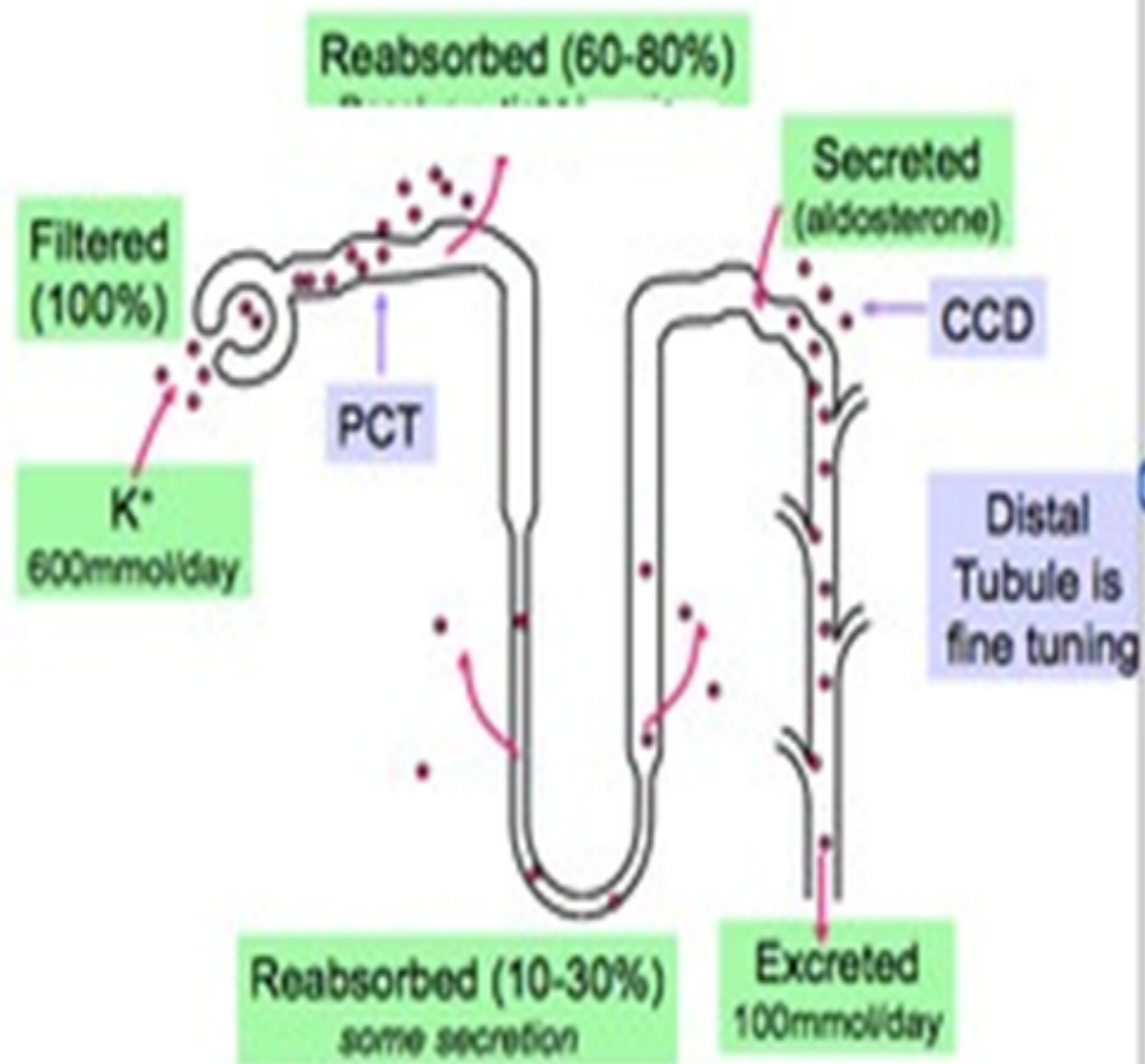
- Reabsorbs 60–80% by an active mechanism.

Loop of henle:

- Reabsorbs about 20% actively in the thick ascending limb by Na-K-2CL co-transporters.

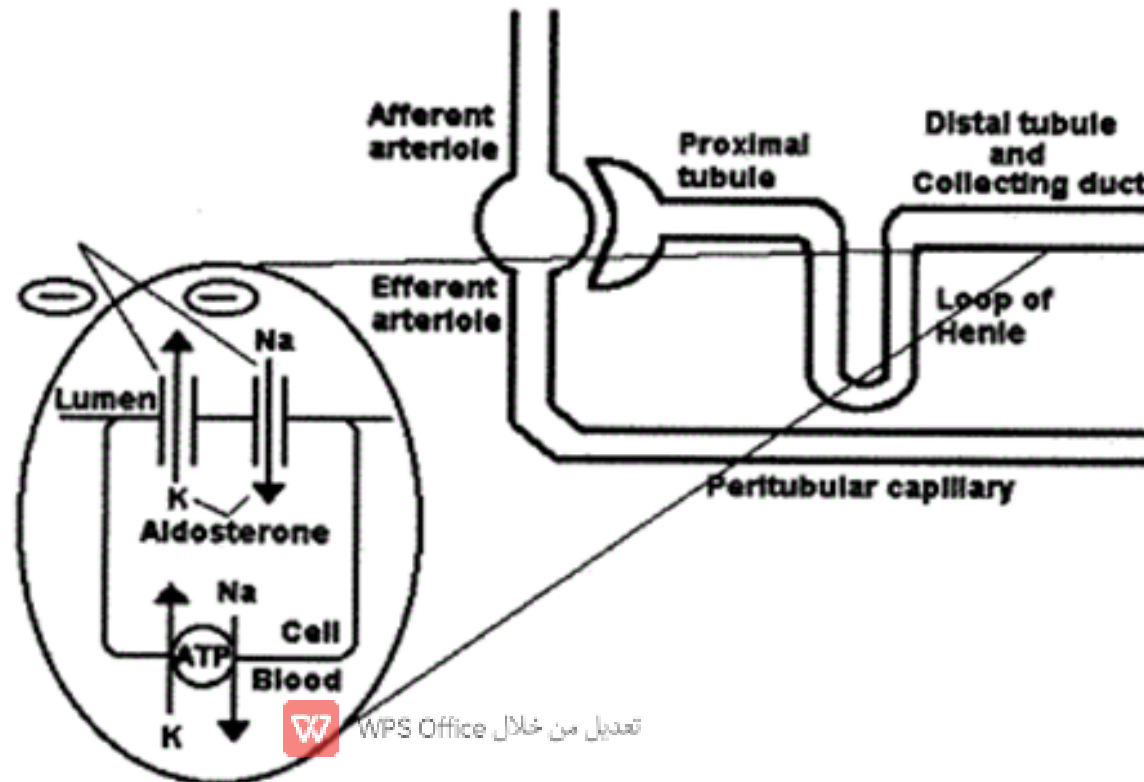
DCT& CDs:

- K secretion by the collecting duct cells.
- Secretion is hormonally dependent (depend on aldosterone)



Mechanism of K secretion

- K secreted down its electrochemical gradient from principal cells to tubular lumen through the apical membranes.
- The chemical gradient created by Na-K ATPase pumps on the basolateral membranes which increase K inside the cells.



Factors affecting K secretion:

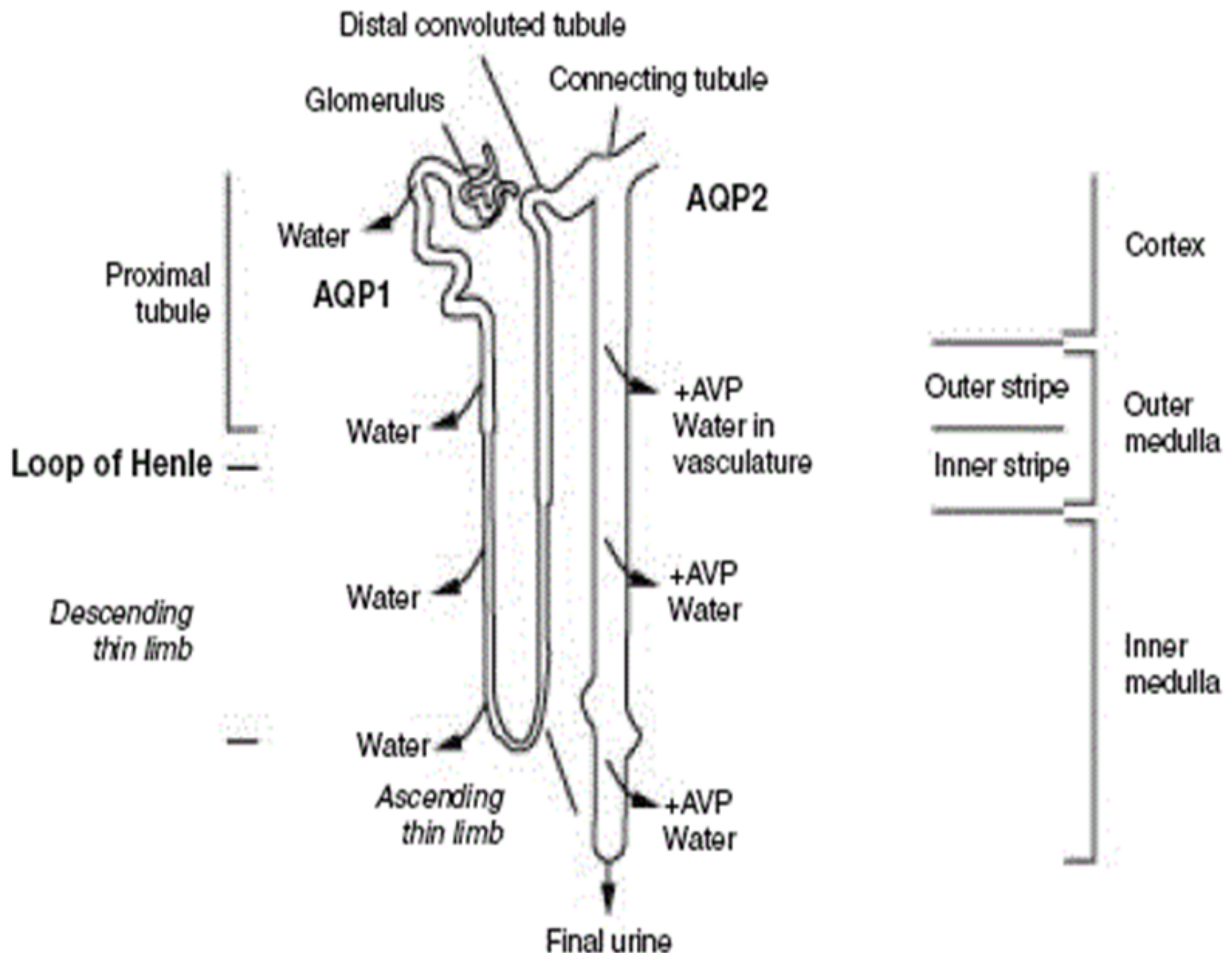
- Dietary K
- Plasma K
- Aldosterone
- Tubular flow rate
- Na concentration in the tubular fluid
- pH
- Diuretics



Water handling

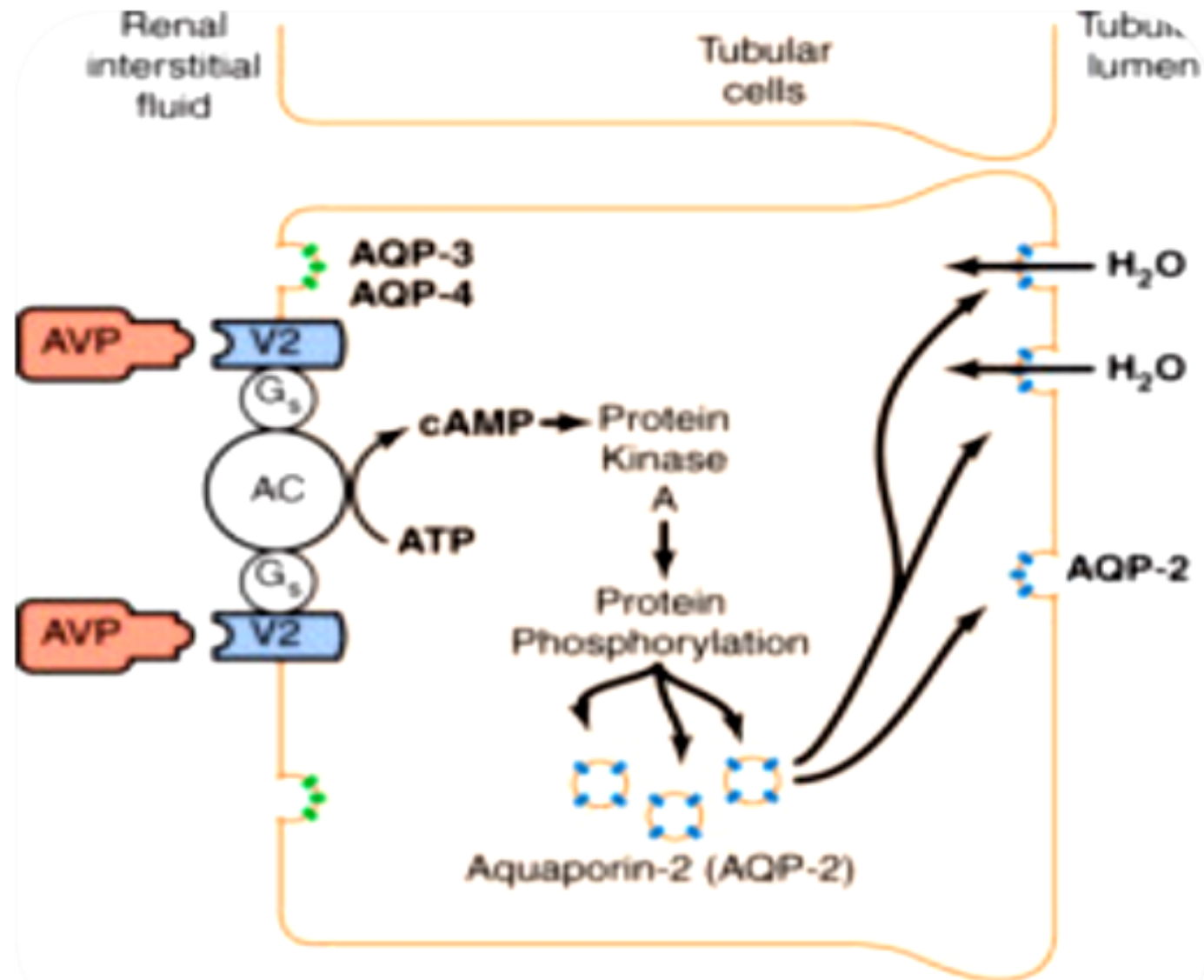
- 180 L/ day of water is filtered through the glomeruli .
- The average urine volume is about 1–1.5 L/ day.
- At least 87% of the filtered water is reabsorbed passively by the renal tubules through special channels (aquaporins).





ADH mechanism

Anti diuretic hormone (Vasopressin) (AVP)



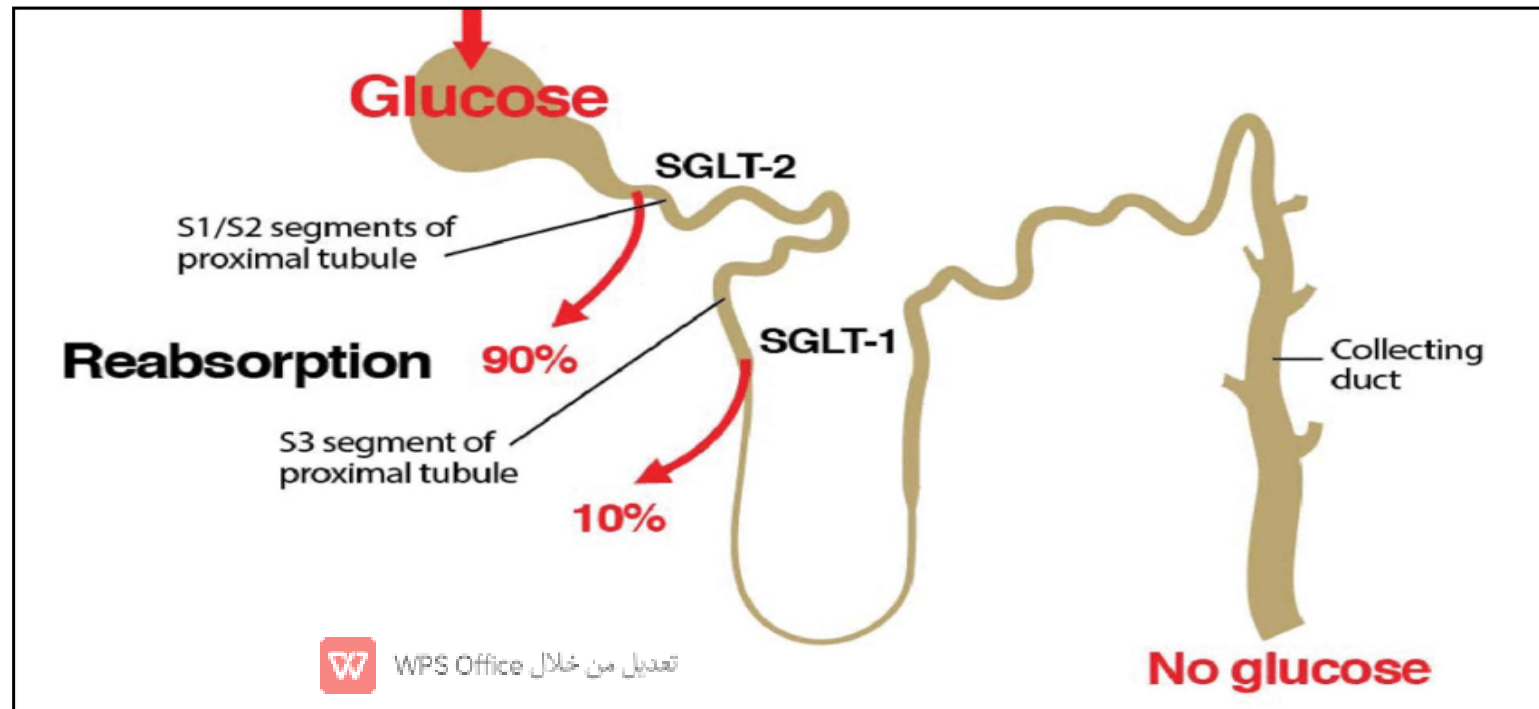
Clinical notes

- High level of ADH in (SIADH), urine is highly concentrated, with 0.5 L/day volume.
- Absence of ADH (Diabetes Insipidus), urine is diluted with 23 L/day volume.
- Two types of DI:
 - 1) nephrogenic : defect in V2 or AQP-2.
 - 2) neurogenic DI: defect in synthesis or release of ADH.



Glucose handling

- The amount filtered depends on plasma glucose level.
- Filtered load = glucose in plasma x GFR
- Reabsorption occurs almost totally in the PCT.
- **T_mG** : is the maximum rate of glucose that can be transported by carriers in PCT per unit time.



- Glucose appear in urine if the TmG is exceeded 375mg/min for men & 300mg/min for women.
- These values never reached when Glucose level is normal.

Renal threshold of glucose:

- Is the plasma level at which all glucose carriers are saturated & the glucose 1st appear in urine.
- The actual threshold is 180mg /dl (in venous blood) & 200 mg /dl (in arterial blood).



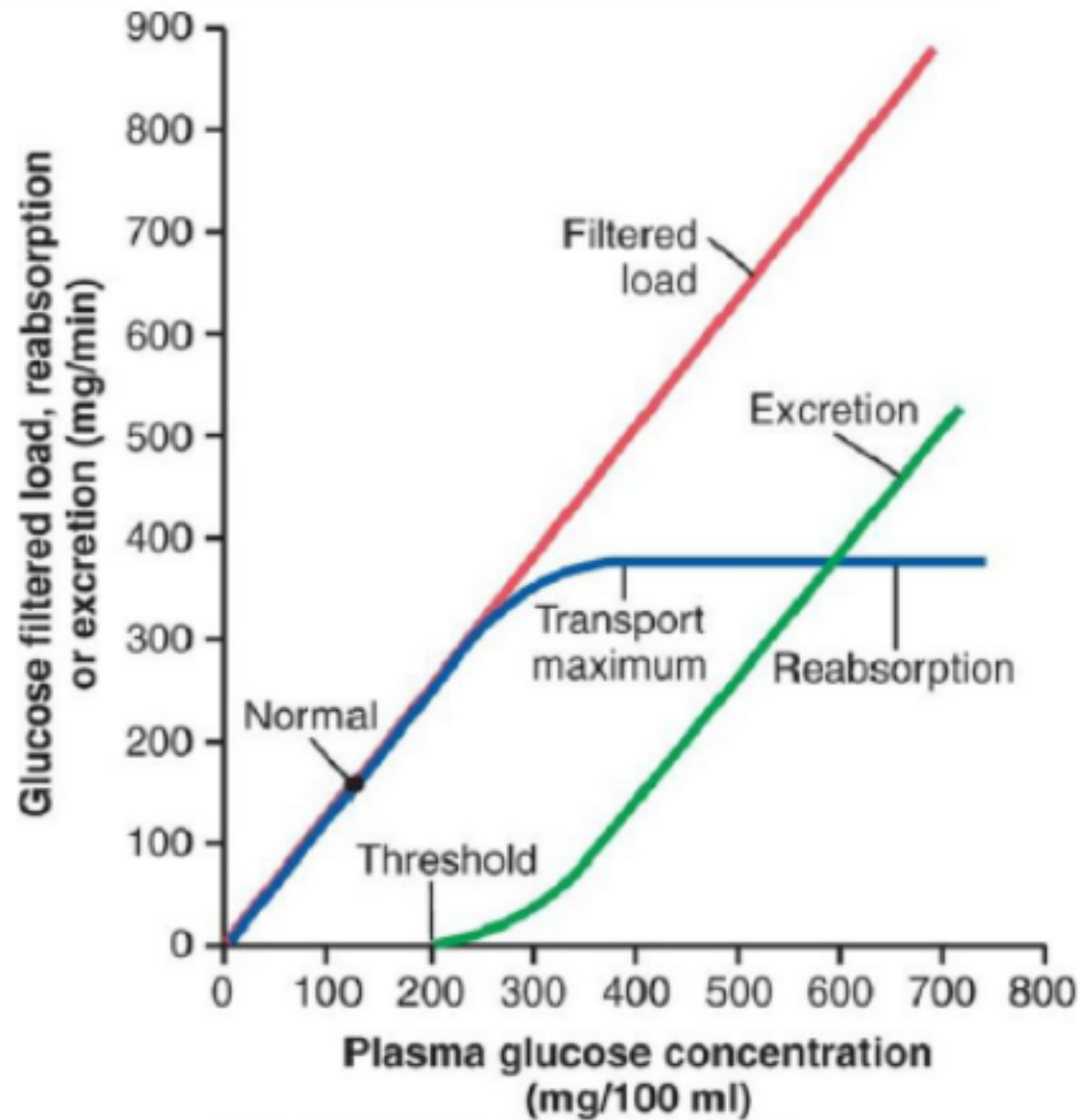
- At this level the filtered load of glucose is equivalent to the T_m (the upper limit of transport). And above this level glucose appears in urine.
- However, when calculating the actual level of glucose in plasma, which gives a filtered load that equals the T_m , it is found to be higher than 180mg\dl.
- Filtered load = GFR x plasma concentration
- Plasma concentration = filtered load (or T_m at this case) / GFR
- Plasma concentration = 375mg/min / 125ml/min



- Plasma concentration = 3 mg/ml = 300 mg/dl.
- The difference between the calculated renal threshold and the true one is related to the fact that Some nephrons reabsorb small amount of glucose whereas others can reabsorb large amount, so the T_m represents the average reabsorptive capacity of all nephrons.

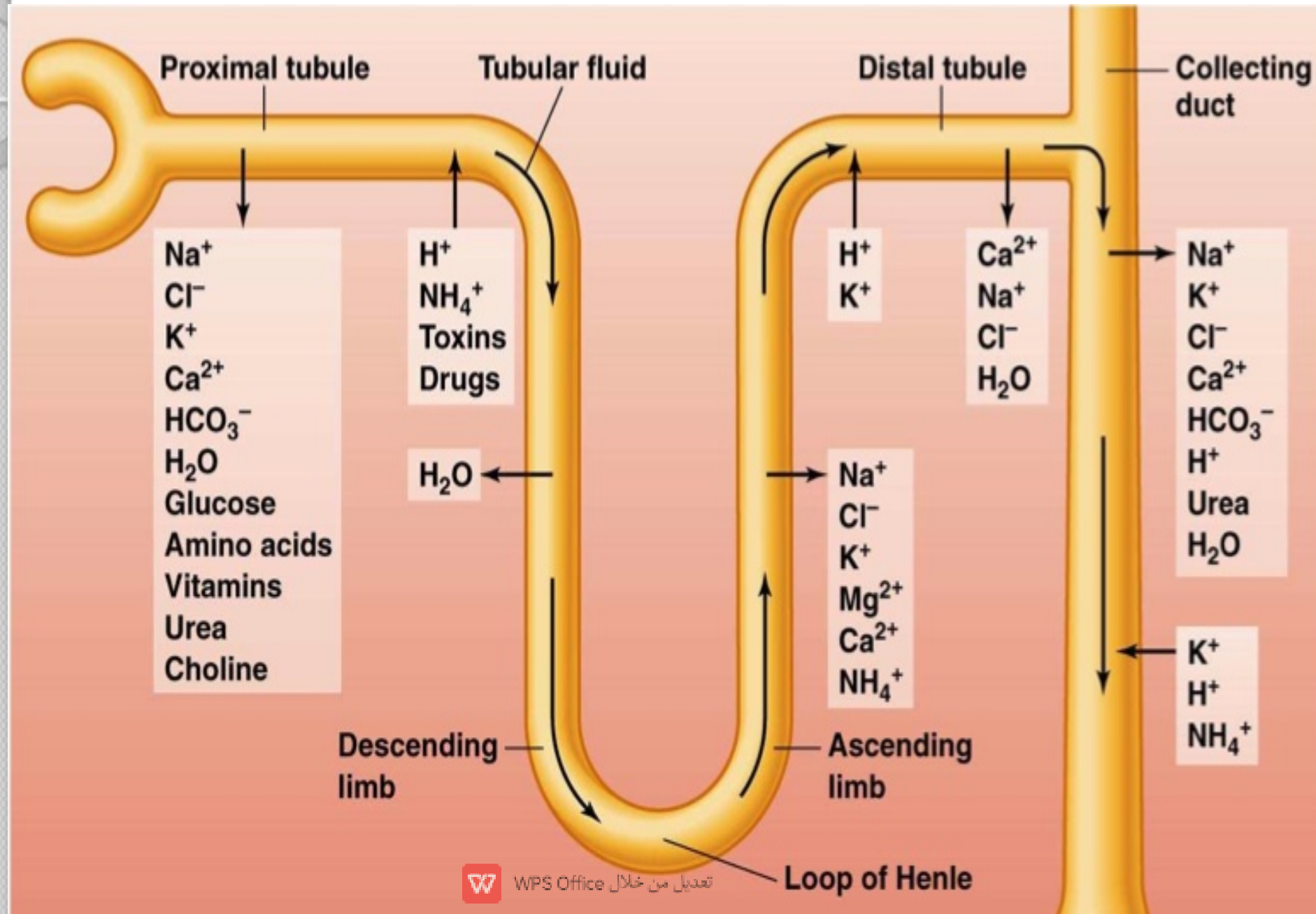


Glucose titration curve



Relations among the filtered load of glucose, the rate of glucose reabsorption by the renal tubules, and the rate of glucose excretion in the urine

Summary



Diuretics

- 1. Sites of drug action
- 2. Osmotic diuretics
- 3. Thiazide diuretics
- 4. Loop diuretics
- 5. Potassium-sparing diuretics



Definitions

Diuretic:

substance that promotes the excretion of urine
caffeine, cranberry juice,
alcohol

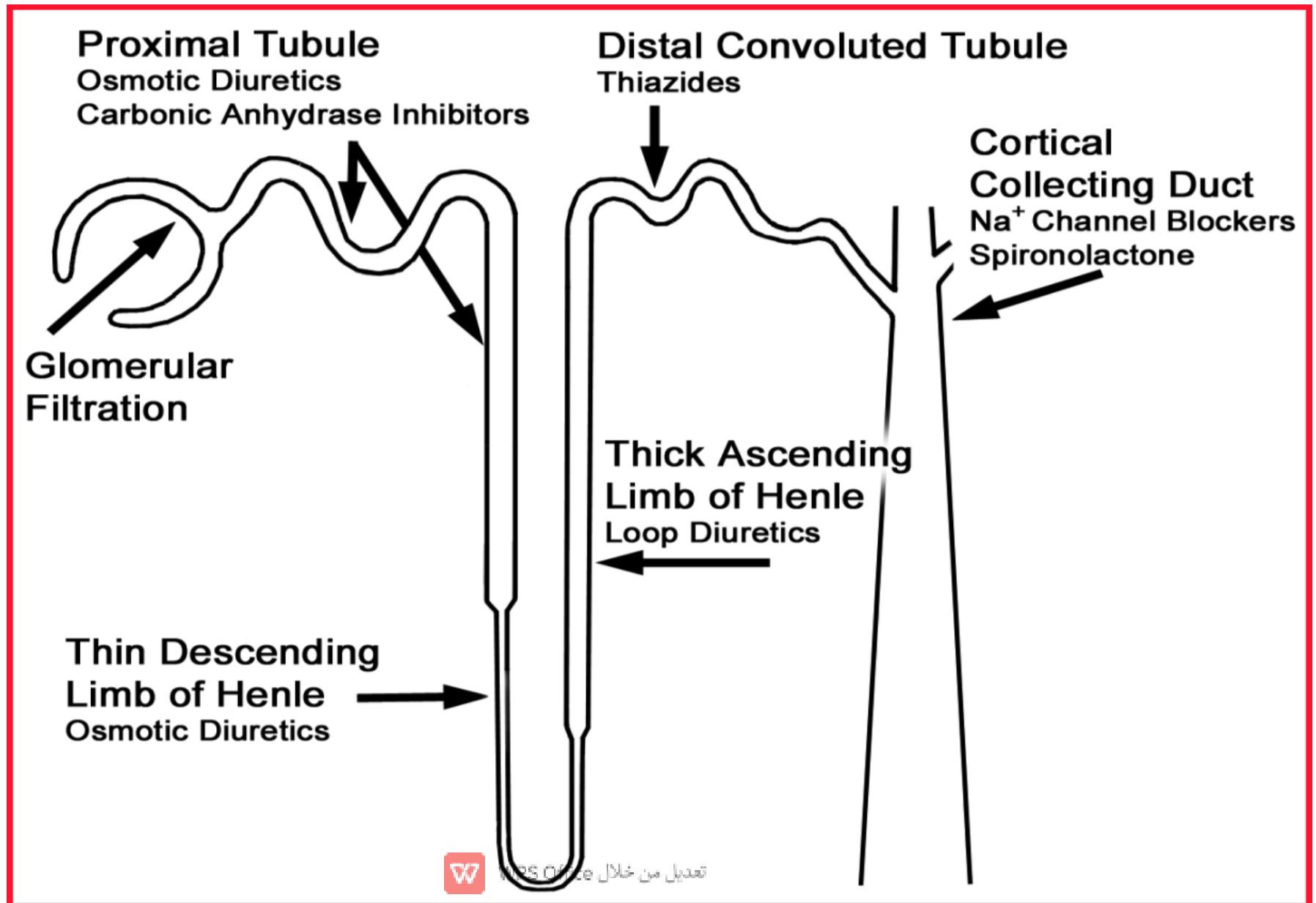


Natriuretic:

substance that promotes the renal excretion of Na^+



diuretics: Sites of Action

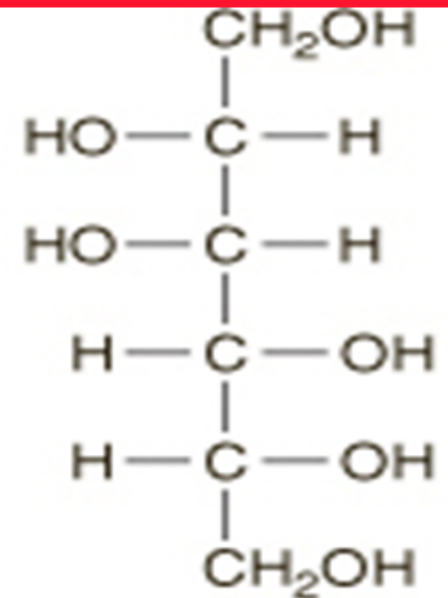


Osmotic Diuretics

do not interact with receptors
or directly block renal
transport

activity dependent on
development of osmotic
pressure

- Mannitol
- Urea
- Glycerol



Mannitol
(Osmotic)

Mechanism of Action

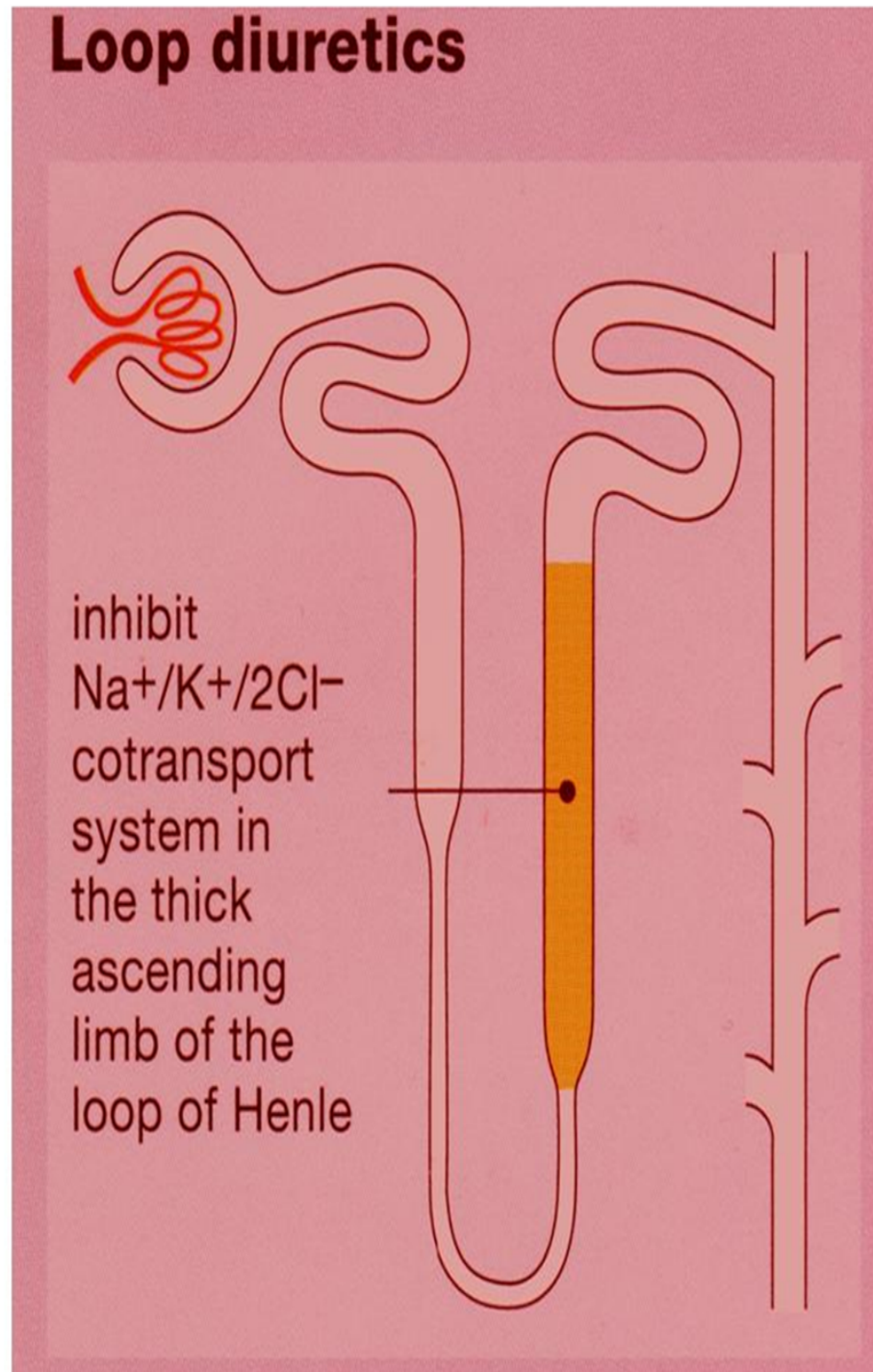
- osmotic diuretics are not reabsorbed
- increases osmotic pressure specifically in the proximal tubule and loop of Henle
- prevents passive reabsorption of H₂O
- osmotic force solute in lumen > osmotic force of reabsorbed Na⁺
- increased H₂O and Na⁺ excretion



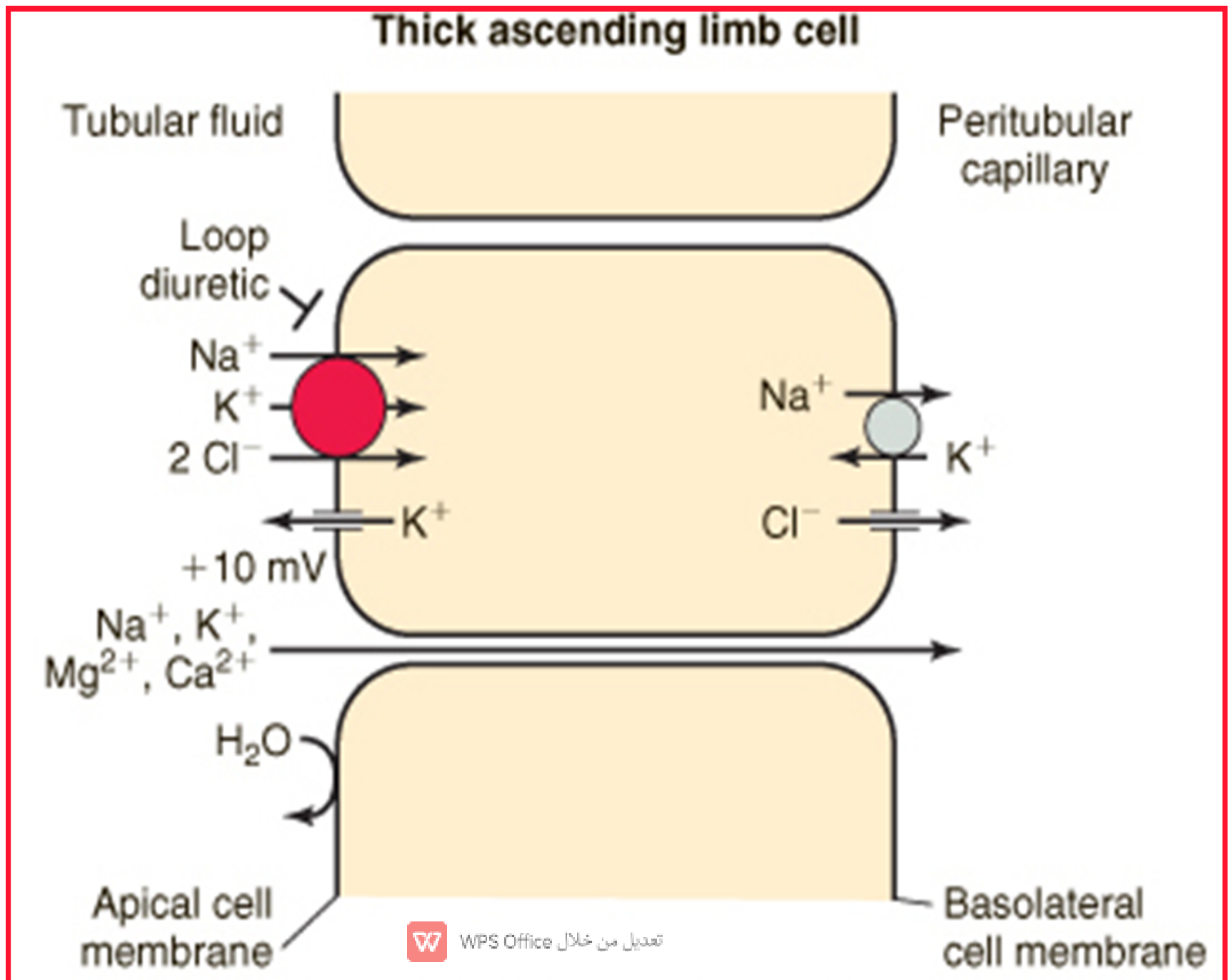
Loop Diuretics

active in “loop” of Henle

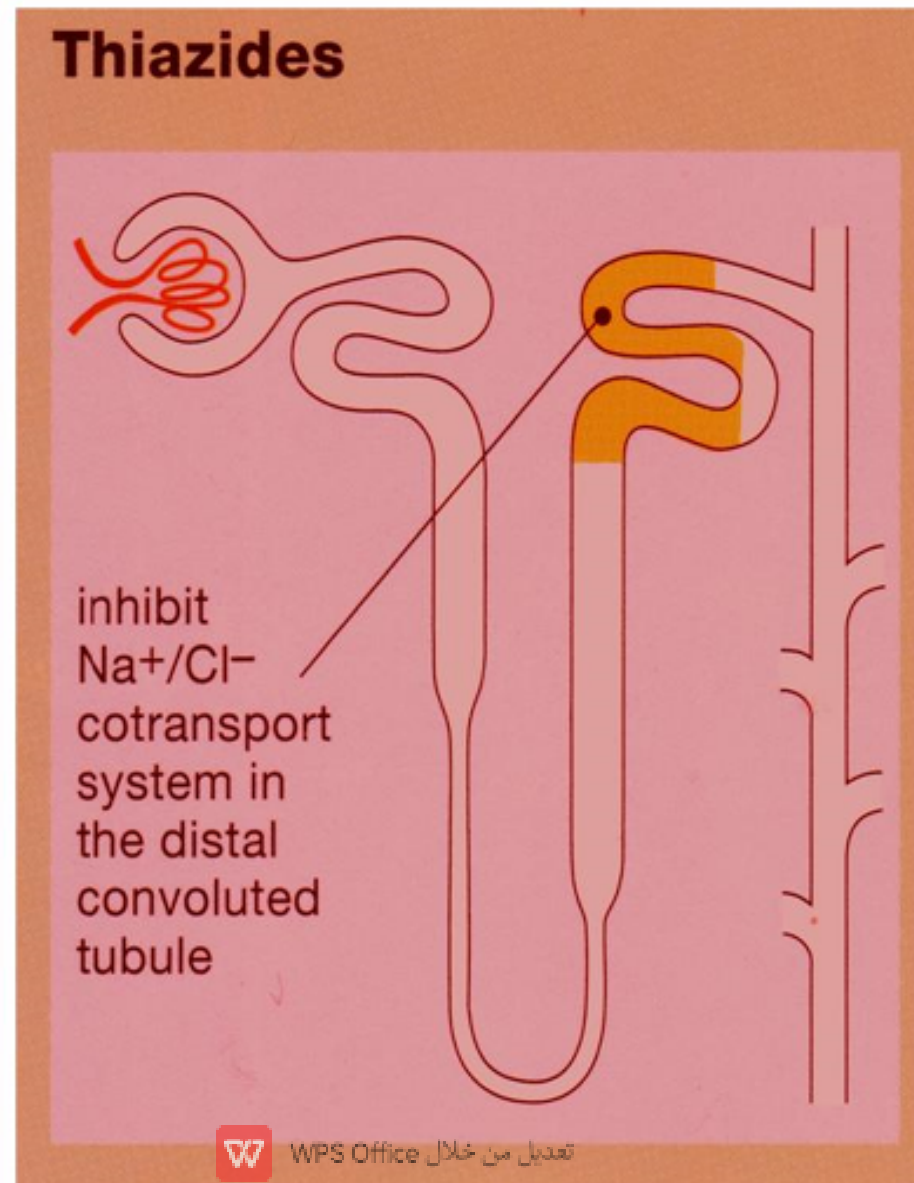
- Furosemide (Lazix)
- Bumetanide



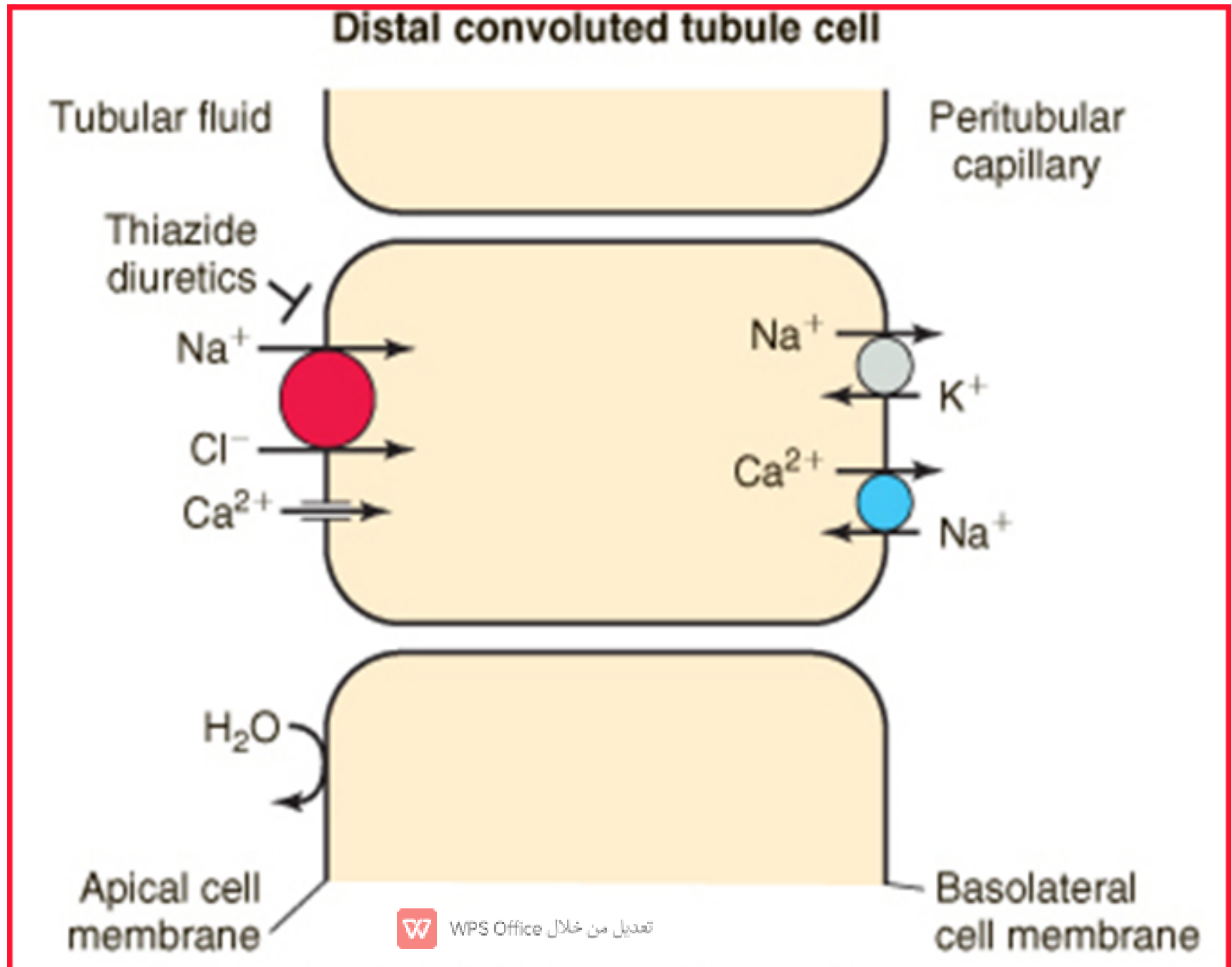
Mechanism of Action



Thiazide Diuretics



Mechanism of Action



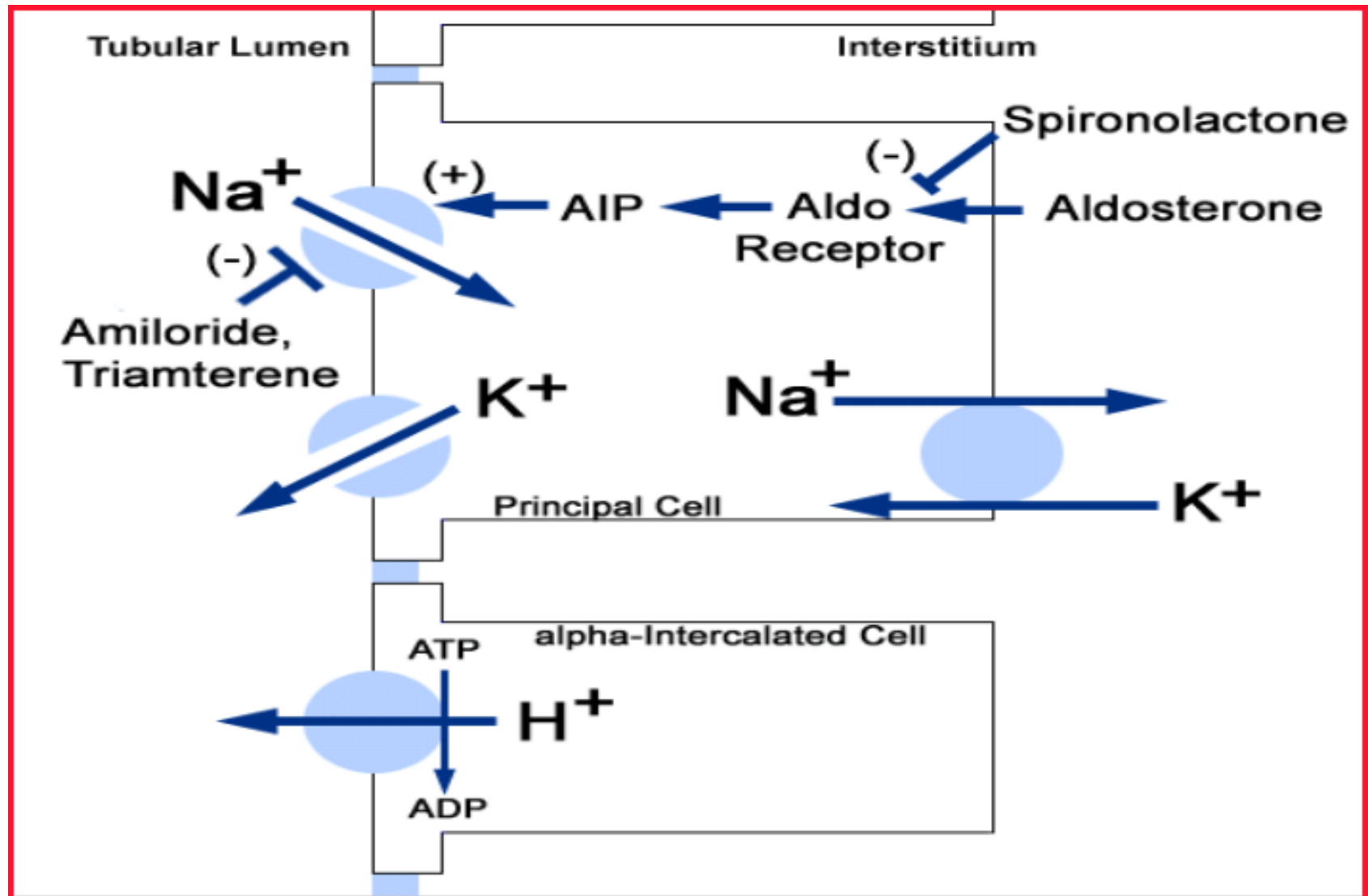
K⁺ sparing diuretics

- Spironolactone: Antagonize the action of aldosterone
- Amiloride: inhibit the ENaCs

Potassium-sparing diuretics



Mechanism of Action







THANK YOU

